

Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)

SHAH

SY-133

SY-133

SY01331

ST1

Design: SY-133

Standard Survey Report

05 November, 2018

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-133
Project:	SHAH	TVD Reference:	RTKB @ 349.4ft
Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Project	SHAH, SHAH		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	Nahrwan - U. Arab Emer.		
Map Zone:	Zone 39N (48 E to 54 E)		

Site	SY-133, Shah-B		
Site Position:		Northing:	2,527,960.04 m
From:	Map	Easting:	781,244.93 m
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in
		Latitude:	22° 50' 15.941 N
		Longitude:	53° 44' 24.996 E
		Grid Convergence:	1.06 °

Well	SY-133		
Well Position	+N/-S	0.0 ft	Northing: 2,527,960.04 m
	+E/-W	0.0 ft	Easting: 781,244.93 m
Position Uncertainty	0.0 ft	Wellhead Elevation:	ft
		Latitude:	22° 50' 15.941 N
		Longitude:	53° 44' 24.996 E
		Ground Level:	319.0 ft

Wellbore	ST1				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	WMM_2005	12/31/2009	1.55	34.86	42,587.91038661

Design	SY-133			
Audit Notes:				
Version:	Phase:	ACTUAL	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	51.88

Survey Program	Date	10/25/2018		
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description
25.0	4,695.0	GYRO SURVEY (ST1)	CT_GYRO_MS	Continuous Gyro Multishot
4,696.0	8,872.0	FINAL SURVEY (ST1)	MWD ISCWSA	MWD - Standard

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00	
25.0	0.08	151.01	25.0	0.0	0.0	0.0	0.32	0.32	0.00	
50.0	0.15	151.01	50.0	-0.1	0.0	0.0	0.28	0.28	0.00	
75.0	0.23	151.01	75.0	-0.1	0.1	0.0	0.32	0.32	0.00	
100.0	0.31	151.01	100.0	-0.2	0.1	0.0	0.32	0.32	0.00	
125.0	0.39	151.01	125.0	-0.4	0.2	-0.1	0.32	0.32	0.00	
150.0	0.46	151.01	150.0	-0.5	0.3	-0.1	0.28	0.28	0.00	
175.0	0.54	151.01	175.0	-0.7	0.4	-0.1	0.32	0.32	0.00	
200.0	0.53	133.90	200.0	-0.9	0.5	-0.1	0.64	-0.04	-68.44	

Survey Report

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Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
225.0	0.61	123.84	225.0	-1.1	0.7	-0.1	0.51	0.32	-40.24
250.0	0.65	100.71	250.0	-1.2	1.0	0.1	1.02	0.16	-92.52
275.0	0.69	87.74	275.0	-1.2	1.3	0.3	0.63	0.16	-51.88
300.0	0.53	57.47	300.0	-1.1	1.5	0.5	1.42	-0.64	-121.08
325.0	0.24	29.52	325.0	-1.0	1.6	0.7	1.35	-1.16	-111.80
350.0	0.20	181.78	350.0	-1.0	1.7	0.7	1.71	-0.16	609.04
375.0	0.41	154.89	375.0	-1.1	1.7	0.6	0.99	0.84	-107.56
400.0	0.54	139.26	400.0	-1.3	1.8	0.6	0.73	0.52	-62.52
425.0	0.61	125.10	425.0	-1.5	2.0	0.7	0.63	0.28	-56.64
450.0	0.65	104.24	450.0	-1.6	2.3	0.8	0.93	0.16	-83.44
475.0	0.65	82.87	475.0	-1.6	2.5	1.0	0.96	0.00	-85.48
500.0	0.60	71.00	500.0	-1.5	2.8	1.3	0.55	-0.20	-47.48
525.0	0.53	59.20	525.0	-1.4	3.0	1.5	0.54	-0.28	-47.20
550.0	0.54	32.29	550.0	-1.3	3.2	1.7	1.00	0.04	-107.64
575.0	0.54	17.24	575.0	-1.1	3.3	1.9	0.57	0.00	-60.20
600.0	0.45	6.88	600.0	-0.8	3.3	2.1	0.51	-0.36	-41.44
625.0	0.34	336.61	625.0	-0.7	3.3	2.2	0.93	-0.44	-121.08
650.0	0.43	351.65	650.0	-0.5	3.3	2.2	0.54	0.36	60.16
675.0	0.35	343.03	675.0	-0.4	3.2	2.3	0.40	-0.32	-34.48
700.0	0.30	2.32	700.0	-0.2	3.2	2.4	0.48	-0.20	77.16
725.0	0.31	33.50	725.0	-0.1	3.2	2.5	0.66	0.04	124.72
750.0	0.39	57.67	750.0	0.0	3.4	2.6	0.66	0.32	96.68
775.0	0.48	63.09	775.0	0.1	3.5	2.8	0.40	0.36	21.68
800.0	0.59	64.50	800.0	0.2	3.7	3.1	0.44	0.44	5.64
825.0	0.71	66.89	825.0	0.3	4.0	3.3	0.49	0.48	9.56
850.0	0.80	57.41	850.0	0.5	4.3	3.7	0.61	0.36	-37.92
875.0	0.86	56.71	875.0	0.7	4.6	4.0	0.24	0.24	-2.80
900.0	0.92	50.51	900.0	0.9	4.9	4.4	0.45	0.24	-24.80
925.0	0.99	47.47	925.0	1.2	5.2	4.8	0.35	0.28	-12.16
950.0	1.09	44.88	950.0	1.5	5.5	5.3	0.44	0.40	-10.36
975.0	1.04	39.40	975.0	1.8	5.8	5.7	0.45	-0.20	-21.92
1,000.0	0.97	37.00	999.9	2.2	6.1	6.2	0.33	-0.28	-9.60
1,025.0	0.90	25.66	1,024.9	2.5	6.3	6.5	0.79	-0.28	-45.36
1,050.0	0.97	26.24	1,049.9	2.9	6.5	6.9	0.28	0.28	2.32
1,075.0	0.93	21.74	1,074.9	3.3	6.7	7.3	0.34	-0.16	-18.00
1,100.0	0.93	23.72	1,099.9	3.6	6.8	7.6	0.13	0.00	7.92
1,125.0	0.90	25.79	1,124.9	4.0	7.0	8.0	0.18	-0.12	8.28
1,150.0	0.92	24.72	1,149.9	4.4	7.2	8.3	0.10	0.08	-4.28
1,175.0	0.89	20.94	1,174.9	4.7	7.3	8.7	0.27	-0.12	-15.12
1,200.0	0.93	24.07	1,199.9	5.1	7.5	9.0	0.26	0.16	12.52
1,225.0	0.94	22.74	1,224.9	5.5	7.6	9.4	0.10	0.04	-5.32
1,250.0	1.02	21.90	1,249.9	5.9	7.8	9.8	0.33	0.32	-3.36

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Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
1,275.0	0.91	19.46	1,274.9	6.3	8.0	10.1	0.47	-0.44	-9.76	
1,300.0	0.91	17.52	1,299.9	6.6	8.1	10.5	0.12	0.00	-7.76	
1,325.0	0.96	12.75	1,324.9	7.0	8.2	10.8	0.37	0.20	-19.08	
1,350.0	1.02	12.52	1,349.9	7.4	8.3	11.1	0.24	0.24	-0.92	
1,375.0	0.99	4.74	1,374.9	7.9	8.3	11.4	0.56	-0.12	-31.12	
1,400.0	1.04	1.67	1,399.9	8.3	8.4	11.7	0.30	0.20	-12.28	
1,425.0	1.04	1.54	1,424.9	8.8	8.4	12.0	0.01	0.00	-0.52	
1,450.0	1.01	355.48	1,449.9	9.2	8.4	12.3	0.45	-0.12	-24.24	
1,475.0	1.06	352.22	1,474.9	9.7	8.3	12.5	0.31	0.20	-13.04	
1,500.0	1.07	349.47	1,499.9	10.1	8.2	12.7	0.21	0.04	-11.00	
1,525.0	1.14	353.12	1,524.9	10.6	8.2	13.0	0.40	0.28	14.60	
1,550.0	1.11	354.55	1,549.9	11.1	8.1	13.2	0.16	-0.12	5.72	
1,575.0	1.11	353.72	1,574.9	11.6	8.1	13.5	0.06	0.00	-3.32	
1,600.0	1.06	352.08	1,599.9	12.0	8.0	13.7	0.24	-0.20	-6.56	
1,625.0	1.02	349.15	1,624.9	12.5	7.9	14.0	0.27	-0.16	-11.72	
1,650.0	1.05	348.99	1,649.9	12.9	7.9	14.2	0.12	0.12	-0.64	
1,675.0	1.03	346.74	1,674.8	13.4	7.8	14.4	0.18	-0.08	-9.00	
1,700.0	1.09	344.00	1,699.8	13.8	7.6	14.5	0.31	0.24	-10.96	
1,725.0	1.08	346.38	1,724.8	14.3	7.5	14.7	0.18	-0.04	9.52	
1,750.0	1.11	345.76	1,749.8	14.7	7.4	14.9	0.13	0.12	-2.48	
1,775.0	0.93	338.19	1,774.8	15.2	7.3	15.1	0.90	-0.72	-30.28	
1,800.0	0.92	326.35	1,799.8	15.5	7.1	15.2	0.76	-0.04	-47.36	
1,825.0	0.84	324.58	1,824.8	15.8	6.9	15.2	0.34	-0.32	-7.08	
1,850.0	0.66	319.13	1,849.8	16.1	6.7	15.2	0.77	-0.72	-21.80	
1,875.0	0.33	327.54	1,874.8	16.3	6.5	15.2	1.35	-1.32	33.64	
1,900.0	0.20	30.62	1,899.8	16.4	6.5	15.2	1.19	-0.52	252.32	
1,925.0	0.43	24.06	1,924.8	16.5	6.6	15.4	0.93	0.92	-26.24	
1,950.0	0.58	39.11	1,949.8	16.7	6.7	15.6	0.80	0.60	60.20	
1,975.0	0.66	25.76	1,974.8	16.9	6.8	15.8	0.66	0.32	-53.40	
2,000.0	0.80	24.30	1,999.8	17.2	7.0	16.1	0.56	0.56	-5.84	
2,025.0	0.87	14.43	2,024.8	17.5	7.1	16.4	0.64	0.28	-39.48	
2,050.0	0.89	13.23	2,049.8	17.9	7.2	16.7	0.11	0.08	-4.80	
2,075.0	0.92	11.22	2,074.8	18.3	7.3	17.0	0.17	0.12	-8.04	
2,100.0	0.85	6.27	2,099.8	18.7	7.3	17.3	0.41	-0.28	-19.80	
2,125.0	0.93	357.70	2,124.8	19.1	7.3	17.5	0.62	0.32	-34.28	
2,150.0	0.95	347.23	2,149.8	19.5	7.3	17.8	0.69	0.08	-41.88	
2,175.0	0.95	340.08	2,174.8	19.9	7.2	17.9	0.47	0.00	-28.60	
2,200.0	0.92	336.38	2,199.8	20.2	7.0	18.0	0.27	-0.12	-14.80	
2,225.0	0.89	335.67	2,224.8	20.6	6.9	18.1	0.13	-0.12	-2.84	
2,250.0	0.83	331.39	2,249.8	20.9	6.7	18.2	0.35	-0.24	-17.12	
2,275.0	0.81	325.71	2,274.8	21.2	6.5	18.2	0.33	-0.08	-22.72	
2,300.0	0.63	321.67	2,299.8	21.5	6.3	18.2	0.75	-0.72	-16.16	
2,325.0	0.47	304.16	2,324.8	21.7	6.2	18.2	0.92	-0.64	-70.04	

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-133
Project:	SHAH	TVD Reference:	RTKB @ 349.4ft
Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
2,350.0	0.33	320.42	2,349.8	21.8	6.0	18.2	0.72	-0.56	65.04
2,375.0	0.32	345.26	2,374.8	21.9	6.0	18.2	0.56	-0.04	99.36
2,400.0	0.46	39.41	2,399.8	22.0	6.0	18.3	1.51	0.56	216.60
2,425.0	0.35	24.08	2,424.8	22.2	6.1	18.5	0.61	-0.44	-61.32
2,450.0	0.51	24.96	2,449.8	22.4	6.2	18.7	0.64	0.64	3.52
2,475.0	0.65	25.91	2,474.8	22.6	6.3	18.9	0.56	0.56	3.80
2,500.0	0.73	24.45	2,499.8	22.9	6.4	19.2	0.33	0.32	-5.84
2,525.0	0.72	21.23	2,524.8	23.2	6.5	19.4	0.17	-0.04	-12.88
2,550.0	0.72	18.80	2,549.8	23.5	6.6	19.7	0.12	0.00	-9.72
2,575.0	0.67	4.52	2,574.8	23.7	6.7	19.9	0.72	-0.20	-57.12
2,600.0	0.67	345.76	2,599.8	24.0	6.7	20.1	0.87	0.00	-75.04
2,625.0	0.70	346.23	2,624.8	24.3	6.6	20.2	0.12	0.12	1.88
2,650.0	0.63	339.45	2,649.8	24.6	6.5	20.3	0.42	-0.28	-27.12
2,675.0	0.67	356.01	2,674.8	24.9	6.5	20.4	0.77	0.16	66.24
2,700.0	0.73	344.14	2,699.8	25.2	6.4	20.6	0.63	0.24	-47.48
2,725.0	0.77	346.87	2,724.8	25.5	6.3	20.7	0.21	0.16	10.92
2,750.0	0.80	345.92	2,749.8	25.8	6.2	20.9	0.13	0.12	-3.80
2,775.0	0.78	353.07	2,774.8	26.2	6.2	21.0	0.40	-0.08	28.60
2,800.0	0.82	325.29	2,799.7	26.5	6.1	21.1	1.54	0.16	-111.12
2,825.0	3.09	306.89	2,824.7	27.0	5.4	21.0	9.31	9.08	-73.60
2,850.0	3.57	343.83	2,849.7	28.2	4.7	21.1	8.63	1.92	147.76
2,875.0	4.63	5.98	2,874.6	29.9	4.6	22.1	7.55	4.24	88.60
2,900.0	5.43	18.07	2,899.5	32.1	5.0	23.7	5.30	3.20	48.36
2,925.0	6.45	31.24	2,924.4	34.4	6.1	26.0	6.78	4.08	52.68
2,950.0	7.43	43.67	2,949.2	36.8	8.0	29.0	7.15	3.92	49.72
2,975.0	8.81	52.27	2,974.0	39.1	10.6	32.5	7.34	5.52	34.40
3,000.0	10.71	56.21	2,998.6	41.6	14.0	36.7	8.05	7.60	15.76
3,025.0	11.63	58.18	3,023.1	44.2	18.1	41.5	3.98	3.68	7.88
3,050.0	11.42	57.98	3,047.6	46.8	22.4	46.5	0.86	-0.84	-0.80
3,075.0	11.38	58.01	3,072.1	49.4	26.5	51.4	0.16	-0.16	0.12
3,100.0	11.46	58.27	3,096.6	52.1	30.7	56.3	0.38	0.32	1.04
3,125.0	11.47	57.85	3,121.1	54.7	35.0	61.3	0.34	0.04	-1.68
3,150.0	11.63	57.85	3,145.6	57.3	39.2	66.2	0.64	0.64	0.00
3,175.0	12.01	57.92	3,170.1	60.1	43.5	71.3	1.52	1.52	0.28
3,200.0	12.86	58.31	3,194.5	62.9	48.1	76.7	3.42	3.40	1.56
3,225.0	14.52	58.76	3,218.8	66.0	53.2	82.6	6.65	6.64	1.80
3,250.0	15.13	59.49	3,243.0	69.3	58.6	88.9	2.55	2.44	2.92
3,275.0	15.09	59.96	3,267.1	72.6	64.3	95.4	0.52	-0.16	1.88
3,300.0	15.55	60.28	3,291.2	75.9	70.0	101.9	1.87	1.84	1.28
3,325.0	16.89	60.23	3,315.2	79.3	76.1	108.8	5.36	5.36	-0.20
3,350.0	18.00	60.14	3,339.1	83.0	82.6	116.2	4.44	4.44	-0.36
3,375.0	18.45	60.86	3,362.8	86.9	89.4	124.0	2.01	1.80	2.88

Survey Report

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Project:	SHAH	TVD Reference:	RTKB @ 349.4ft
Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
3,400.0	19.15	62.31	3,386.5	90.7	96.5	131.9	3.37	2.80	5.80
3,425.0	20.96	62.46	3,410.0	94.7	104.1	140.3	7.24	7.24	0.60
3,450.0	21.82	62.72	3,433.2	98.9	112.2	149.3	3.46	3.44	1.04
3,475.0	22.55	62.89	3,456.4	103.2	120.5	158.6	2.93	2.92	0.68
3,500.0	24.45	62.27	3,479.3	107.8	129.4	168.3	7.66	7.60	-2.48
3,525.0	26.33	62.13	3,501.9	112.8	138.9	178.9	7.52	7.52	-0.56
3,550.0	26.87	62.34	3,524.3	118.0	148.8	189.9	2.19	2.16	0.84
3,575.0	27.20	61.40	3,546.5	123.4	158.8	201.1	2.16	1.32	-3.76
3,600.0	28.33	60.76	3,568.7	129.0	169.0	212.6	4.67	4.52	-2.56
3,625.0	29.61	59.87	3,590.5	135.0	179.5	224.6	5.40	5.12	-3.56
3,650.0	30.31	59.87	3,612.2	141.3	190.3	236.9	2.80	2.80	0.00
3,675.0	32.02	58.96	3,633.6	147.9	201.4	249.8	7.09	6.84	-3.64
3,700.0	34.02	57.57	3,654.5	155.0	213.0	263.3	8.55	8.00	-5.56
3,725.0	35.11	57.02	3,675.1	162.7	225.0	277.4	4.54	4.36	-2.20
3,750.0	36.10	56.99	3,695.4	170.6	237.2	291.9	3.96	3.96	-0.12
3,775.0	38.41	57.30	3,715.3	178.8	249.9	307.0	9.27	9.24	1.24
3,800.0	39.93	57.74	3,734.7	187.3	263.2	322.7	6.18	6.08	1.76
3,825.0	40.22	57.79	3,753.9	195.9	276.8	338.7	1.17	1.16	0.20
3,850.0	40.55	56.24	3,772.9	204.7	290.4	354.8	4.23	1.32	-6.20
3,875.0	40.80	55.04	3,791.9	213.9	303.9	371.1	3.28	1.00	-4.80
3,900.0	40.90	54.75	3,810.8	223.3	317.2	387.4	0.86	0.40	-1.16
3,925.0	40.79	54.66	3,829.7	232.8	330.6	403.8	0.50	-0.44	-0.36
3,950.0	40.69	54.53	3,848.6	242.2	343.9	420.1	0.52	-0.40	-0.52
3,975.0	42.15	54.35	3,867.4	251.8	357.3	436.6	5.86	5.84	-0.72
4,000.0	43.17	54.48	3,885.8	261.7	371.1	453.5	4.10	4.08	0.52
4,025.0	43.82	54.59	3,903.9	271.7	385.1	470.7	2.62	2.60	0.44
4,050.0	45.44	54.95	3,921.7	281.8	399.5	488.2	6.56	6.48	1.44
4,075.0	47.66	55.33	3,938.9	292.2	414.4	506.3	8.95	8.88	1.52
4,100.0	48.82	55.64	3,955.5	302.7	429.7	525.0	4.73	4.64	1.24
4,125.0	49.90	55.87	3,971.8	313.4	445.4	543.9	4.38	4.32	0.92
4,150.0	51.89	55.77	3,987.6	324.3	461.5	563.2	7.97	7.96	-0.40
4,175.0	53.85	55.32	4,002.7	335.6	477.9	583.1	7.97	7.84	-1.80
4,200.0	54.63	55.02	4,017.3	347.2	494.5	603.4	3.27	3.12	-1.20
4,225.0	54.83	54.63	4,031.7	358.9	511.2	623.8	1.50	0.80	-1.56
4,250.0	56.01	53.20	4,045.9	371.1	527.9	644.3	6.67	4.72	-5.72
4,275.0	57.31	51.64	4,059.6	383.8	544.4	665.2	7.36	5.20	-6.24
4,300.0	58.10	51.19	4,073.0	397.0	560.9	686.3	3.51	3.16	-1.80
4,325.0	59.44	50.42	4,086.0	410.5	577.5	707.7	5.97	5.36	-3.08
4,350.0	61.86	49.94	4,098.2	424.4	594.2	729.5	9.82	9.68	-1.92
4,375.0	63.46	49.77	4,109.7	438.8	611.2	751.7	6.43	6.40	-0.68
4,400.0	63.65	49.49	4,120.8	453.3	628.3	774.1	1.26	0.76	-1.12
4,425.0	64.14	49.36	4,131.8	467.9	645.3	796.5	2.01	1.96	-0.52
4,450.0	66.06	49.84	4,142.4	482.6	662.6	819.1	7.87	7.68	1.92

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-133
Project:	SHAH	TVD Reference:	RTKB @ 349.4ft
Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,475.0	67.87	49.74	4,152.1	497.4	680.1	842.1	7.25	7.24	-0.40
4,500.0	68.47	49.96	4,161.4	512.4	697.9	865.3	2.54	2.40	0.88
4,525.0	70.18	50.69	4,170.3	527.3	715.9	888.7	7.37	6.84	2.92
4,550.0	72.35	51.45	4,178.3	542.2	734.3	912.4	9.14	8.68	3.04
4,575.0	73.25	51.55	4,185.7	557.0	753.0	936.3	3.62	3.60	0.40
4,600.0	74.70	51.62	4,192.6	572.0	771.8	960.3	5.81	5.80	0.28
4,625.0	76.97	51.74	4,198.7	587.0	790.8	984.5	9.09	9.08	0.48
4,650.0	79.24	52.15	4,203.9	602.1	810.1	1,009.0	9.22	9.08	1.64
4,675.0	79.84	53.02	4,208.4	617.0	829.6	1,033.6	4.18	2.40	3.48
4,695.0	79.74	53.60	4,211.9	628.8	845.4	1,053.2	2.90	-0.50	2.90
4,708.5	80.17	54.40	4,214.3	636.6	856.2	1,066.6	6.64	3.18	5.92
4,830.4	86.72	54.36	4,228.2	707.1	954.6	1,187.5	5.37	5.37	-0.03
4,869.0	87.77	53.73	4,230.1	729.7	985.8	1,226.0	3.17	2.72	-1.63
PRODUCTION LINER									
4,938.4	89.66	52.60	4,231.6	771.3	1,041.3	1,295.4	3.17	2.72	-1.63
5,032.0	89.70	48.76	4,232.1	830.6	1,113.7	1,388.9	4.10	0.04	-4.10
5,125.4	89.73	51.93	4,232.6	890.2	1,185.6	1,482.2	3.39	0.03	3.39
5,218.7	89.18	53.48	4,233.5	946.7	1,259.8	1,575.6	1.76	-0.59	1.66
5,312.2	89.76	52.97	4,234.4	1,002.7	1,334.7	1,669.0	0.83	0.62	-0.55
5,403.6	90.35	51.41	4,234.3	1,058.8	1,406.9	1,760.5	1.82	0.65	-1.71
5,497.3	89.92	50.98	4,234.0	1,117.5	1,479.9	1,854.1	0.65	-0.46	-0.46
5,591.8	89.18	52.69	4,234.8	1,175.9	1,554.2	1,948.6	1.97	-0.78	1.81
5,684.2	89.76	52.13	4,235.6	1,232.2	1,627.5	2,041.0	0.87	0.63	-0.61
5,779.1	89.73	50.92	4,236.1	1,291.3	1,701.7	2,135.9	1.28	-0.03	-1.28
5,872.6	89.95	54.05	4,236.3	1,348.2	1,775.9	2,229.4	3.36	0.24	3.35
5,965.0	90.13	55.50	4,236.3	1,401.5	1,851.4	2,321.6	1.58	0.19	1.57
6,058.3	89.86	54.54	4,236.3	1,455.0	1,927.8	2,414.8	1.07	-0.29	-1.03
6,152.1	89.49	51.12	4,236.8	1,511.6	2,002.5	2,508.6	3.67	-0.39	-3.65
6,246.7	89.43	49.35	4,237.7	1,572.2	2,075.3	2,603.2	1.87	-0.06	-1.87
6,339.4	89.43	48.79	4,238.6	1,632.9	2,145.3	2,695.7	0.60	0.00	-0.60
6,432.4	89.46	47.16	4,239.5	1,695.1	2,214.4	2,788.5	1.75	0.03	-1.75
6,526.3	89.76	45.39	4,240.2	1,760.0	2,282.2	2,881.9	1.91	0.32	-1.89
6,618.9	89.55	44.85	4,240.7	1,825.3	2,347.8	2,973.8	0.63	-0.23	-0.58
6,713.0	88.99	47.52	4,241.9	1,890.5	2,415.7	3,067.5	2.90	-0.59	2.84
6,807.9	89.86	49.80	4,242.9	1,953.1	2,486.9	3,162.2	2.57	0.92	2.40
6,916.3	89.43	51.90	4,243.5	2,021.6	2,571.0	3,270.6	1.98	-0.40	1.94
7,008.5	89.76	52.00	4,244.2	2,078.4	2,643.6	3,362.8	0.37	0.36	0.11
7,097.9	89.58	53.62	4,244.7	2,132.5	2,714.8	3,452.2	1.82	-0.20	1.81
7,190.8	89.55	52.48	4,245.4	2,188.3	2,789.0	3,545.0	1.23	-0.03	-1.23
7,283.2	90.01	52.92	4,245.8	2,244.3	2,862.6	3,637.5	0.69	0.50	0.48
7,376.9	89.52	52.39	4,246.1	2,301.1	2,937.0	3,731.1	0.77	-0.52	-0.57
7,470.7	89.39	51.45	4,247.0	2,359.0	3,010.9	3,824.9	1.01	-0.14	-1.00

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-133
Project:	SHAH	TVD Reference:	RTKB @ 349.4ft
Site:	SY-133	MD Reference:	RTKB @ 349.4ft
Well:	SY-133	North Reference:	Grid
Wellbore:	ST1	Survey Calculation Method:	Minimum Curvature
Design:	SY-133	Database:	EDMPRODR5K

Survey										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	
7,563.9	89.83	52.39	4,247.7	2,416.4	3,084.2	3,918.1	1.11	0.47	1.01	
7,657.3	89.73	52.41	4,248.0	2,473.4	3,158.2	4,011.5	0.11	-0.11	0.02	
7,748.9	89.86	49.98	4,248.4	2,530.9	3,229.7	4,103.2	2.65	0.14	-2.65	
7,845.0	89.83	50.59	4,248.6	2,592.3	3,303.6	4,199.2	0.64	-0.03	0.63	
7,938.8	90.01	53.50	4,248.7	2,649.9	3,377.5	4,292.9	3.11	0.19	3.10	
8,030.4	89.52	53.05	4,249.1	2,704.7	3,450.9	4,384.6	0.73	-0.53	-0.49	
8,125.0	89.21	54.55	4,250.2	2,760.6	3,527.3	4,479.1	1.62	-0.33	1.59	
8,217.9	89.70	51.77	4,251.1	2,816.3	3,601.6	4,572.0	3.04	0.53	-2.99	
8,312.8	89.83	50.79	4,251.4	2,875.6	3,675.6	4,666.8	1.04	0.14	-1.03	
8,406.0	89.67	50.38	4,251.9	2,934.8	3,747.6	4,760.0	0.47	-0.17	-0.44	
8,499.5	89.86	51.26	4,252.2	2,993.9	3,820.1	4,853.5	0.96	0.20	0.94	
8,592.9	89.73	50.52	4,252.6	3,052.8	3,892.5	4,946.8	0.80	-0.14	-0.79	
8,685.4	89.67	52.08	4,253.1	3,110.7	3,964.8	5,039.4	1.69	-0.06	1.68	
8,778.7	90.35	51.51	4,253.0	3,168.4	4,038.1	5,132.7	0.95	0.73	-0.61	
8,843.2	89.36	51.57	4,253.2	3,208.5	4,088.6	5,197.2	1.54	-1.53	0.09	
8,872.0	89.40	52.00	4,253.5	3,226.3	4,111.2	5,226.0	1.50	0.14	1.50	

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
4,869.0	4,230.1	PRODUCTION LINER	7.000	8.500	

Checked By: _____	Approved By: _____	Date: _____
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